

LSU 4.9 analog front-end — topology after rusEFI/FOME wideband module rev C (MIT). See FRONTEND.md

Analog reference VDDA = 3.3 V

Nernst sense: $Un_{sense} = 2.7 \cdot (Un - VM)$, target $Un - VM = 0.45$ V

Virtual ground VM = VDDA/2

Pump driver: Howland current source, $I = -(R33/R31) \cdot (Ip_{dac} - VM)/R\Phi$ Pump current sense: $Ip_{sense} = VM + 10 \cdot (V_{Ip} - V_{Rtrim})$, shunt 61.9 ohm

Heater switch

LSU 4.9 connector (pin order to be verified against datasheet) (analog 0–3.3 V)

Power input: fuse, reverse-polarity P-MOSFET, TVS

CAN (rusEFI wideband protocol)

Interface to MCU board (digital)

5 V buck (TPS54202; pinout and values from memory — VERIFY against datasheet) 3.3 V LDO

0–5 V AFR output: 10..20 AFR = 0.5 V, DAC 0–3.3 V x1.51

Draft — NOT verified by ERC/simulation/bench
Topology after rusEFI/FOME wideband module rev C (MIT)

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Title: LSU 4.9 analog front-end

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